

Chinmay Joshi

✉ chinmay.joshi23@outlook.com ☎ +49 1788266100

🌐 chinmayj23.github.io 🌐 chinmay-joshi23



PROFILE

I am a **machine learning-oriented data scientist** with experience across **applied ML, training, fine-tuning, and evaluating neural networks, analytics, and computer vision**, backed by a strong foundation in **statistics**. I am interested in building **data-centric, reliable solutions** with a practical impact.

EDUCATION

Universität des Saarlandes

Saarbrücken, Germany

Master's in Computer Science (GPA: 1.7)

Oct 2022 – Mar 2026

- Master's Thesis: **On the effect of Data Pruning on Memorization**
(Advisor: Dr. Rebekka Burkholz – CISPA Helmholtz Center for Information Security)
 - Investigating the impact of **Data Pruning on Memorization in Neural Networks**, with a comparative **empirical analysis of training behaviour** across different memorization definitions, with experiments ranging from CIFAR + Resnets to transfer learning of a pretrained ViT on Imagenet to Flowers and Pets.
 - Implemented **large-scale experiments** on A100s using SLURM, eg, estimating stability-based memorization score (Lukasik et al.) with 200 training runs per experiment.
- Recipient of **SaarlandStipendium** and **Santander Scholarship**.

Narsee Monjee Institute of Management Studies

Mumbai, India

Bachelors in Data Science (GPA: 3.8/4)

Aug 2018 – May 2022

- Additional Certification: Financial Markets ([certificate link](#))
- Elective Courses: Cloud Computing (AWS), Financial Engineering and Risk Management

EXPERIENCE

Research Assistant

Dec 2025 – Present

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

Kaiserslautern, Germany

- Developing a modular **Subgroup Discovery engine** to identify statistically interesting patterns in static as well as **time-dependent datasets**. ([Source Code](#))
- Engineering a hybrid approach combining **clustering-based candidate generation, KL divergence scoring, and interpretable rule-based explanations** (via Random Forests) to automatically detect and explain meaningful data anomalies.

Research Assistant

Mar 2025 – Sep 2025

Fraunhofer IZFP – Proj. [Urbanist](#)

Saarbrücken, Germany

- Analyzed **environmental and mobility sensor data** to generate actionable insights on how **transport choices affect environmental impact**, directly supporting sustainability-focused decision-making.
- Modeled raw sensor signals into **interpretable metrics** (UV index, sun exposure, noise levels) to enable real-time evaluation of route sustainability.
- Developed **geospatial routing algorithms** using **OpenStreetMap** to visualize environment-aware routes for a gamified urban mobility application.

Research Intern

Apr 2024 – Aug 2024

National Institute of Informatics

Tokyo, Japan

- Created a **synthetic 3D relighting dataset** using **Blender**, Objaverse assets, and PolyHaven environments; generating **150K+ multi-view, multi-light samples** with corresponding **depth maps and surface normals**.

- Explored **single-image 3D reconstruction with relighting** by conditioning environment lighting on a **diffusion-based prior (Zero-1-to-3)**, analyzing challenges in illumination and view consistency due to the ill-posed nature of the task.

Research Assistant

Fraunhofer IZFP – Proj. [SmartPigHome](#)

Aug 2023 – Mar 2024

Saarbrücken, Germany

- Built a lightweight **audio annotation tool** that replaced a manual paper-based workflow, allowing veterinarians to label data efficiently and **reducing labeling time by up to 80%**.
- Supported computer vision experiments involving **object detection and optical flow** for tracking livestock in video data.

Research Assistant

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) – Proj. [KIMonoS](#)

Jun 2023 – Mar 2024

Saarbrücken, Germany

- Developed an **algorithmic scheduling system** for multi-vehicle shuttle routing, optimizing **passenger wait times and seat utilization** using Python.
- Evaluated system performance under a **12-hour, 60-passenger stochastic simulation** in SUMO to assess behavior under realistic demand patterns.

Computer Vision Intern

Aurify Systems Pvt. Ltd.

Jan 2022 – Apr 2022

Mumbai, India

- Developed **computer vision pipelines** for retail and warehouse analytics, including **lightweight CNN-based object detection** and **template matching with non-maxima suppression** for **bin detection and empty-shelf monitoring**; solutions were showcased in a public exhibition.
- Built a **pose-based activity detection system** using **AlphaPose** and a **classical ML classifier** (e.g., Random Forest) to detect suspicious activities, achieving **98% accuracy** in controlled field tests.
- Prototyped a **CCTV video analytics tool**, combining **image processing, pose estimation, and object detection** to identify workers using mobile phones through a **multi-stage decision pipeline**.

Market Research Analyst

Finbits India

Aug 2020 – Sep 2021

Mumbai, India

- Applied **data-driven financial analysis** on company fundamentals and valuation ratios to assess stock attractiveness, and translated quantitative insights into **written investment commentaries**.

PROJECTS AND PUBLICATIONS

- **Algorithmic Trading Strategy** (IEEE I2CT 2022): Presented **PCA + LSTM** for **forecasting trade signals** using a multicollinear set of 84 technical indicators as features, and leveraging **dimension reduction** for an improved **Time-Series** modeling. Complemented with portfolio optimization strategies, we attain **20% more returns than the Buy-and-Hold benchmark**. ([Paper](#)) ([Source Code](#))
- **Hierarchical VAE for Image Compression**: Implemented a **hierarchical variant of an existing VAE-based image compression model** (Balle et al.), achieving **+5 PSNR over JPEG** at similar bitrate through architectural modifications. ([Report](#))
- **Bird Audio Classification**: Developed an **MFCC-based audio classifier** to identify bird species from sound recordings, achieving **82% test accuracy**.

SKILLS

Programming Languages: Python, SQL, R

Tooling & Frameworks: PyTorch, Tensorflow, Hugging Face, Tableau, Weights & Biases, Git (Version Control), Docker, OpenStreetMap, SLURM, FastAPI, Langchain

Conceptual Understanding: CNNs, LSTMs, Transformers, Diffusion Models, Large Language Models (LLMs), RAGs, Causal Inference, Exploratory Data Analysis, Pattern Mining, Signal Processing, Camera Geometry, 3D Computer Vision, Image Analysis, Time-Series Analysis, Statistics

Languages: English (Native), German (A2), Hindi (Native), Marathi (Native)